

Cassandra Marcussen

Contact	Email Address: cmarcussen@g.harvard.edu Website: https://cassmarcussen.github.io/
Research Interests	My research is in theoretical computer science and probability, with a focus on sub-linear algorithms, random graphs, property testing, and average-case complexity.
Education	Harvard University2022 - Present Ph.D. student in the Theory of Computation group Advisor: Professor Madhu Sudan Columbia University in the City of New York2018 - 2022 B.A. in Mathematics and Computer Science <i>Magna Cum Laude</i>. GPA 4.02. John Jay Scholar
Publications	All author names are listed alphabetically unless noted with *. k-Coloring Graphs in Sublinear Average Time, <i>with Edward Pyne, Ronitt Rubinfeld, Asaf Shapira, and Shlomo Tauber</i>, FOCS (2026), https://arxiv.org/pdf/2607.26592 Quality control in sublinear time: a case study via random graphs, <i>with Ronitt Rubinfeld and Madhu Sudan</i>, FOCS (2026), https://arxiv.org/pdf/2508.16531 Learning and Testing Convex Functions, <i>with Renato Ferreira Pinto Jr., Elchanan Mossel, and Shivam Nadimpalli</i>, RANDOM (2026), https://www.arxiv.org/pdf/2511.11498 Finding the Root in Random Nearest Neighbor Trees, <i>with Anna Brandenberger, Elchanan Mossel, and Madhu Sudan</i>, Random Structures & Algorithms (2026), https://arxiv.org/pdf/2411.14336. Scramble number and tree-cut decompositions, <i>with Lisa Cenek, Lizzie Ferguson, Eyobel Gebre, Jason Meintjes, Ralph Morrison, Liz Ostermeyer, Shefali Ramakrishna, and Ben Weber</i>, The Art of Discrete and Applied Mathematics (2026), https://arxiv.org/pdf/2209.01459.pdf. A Fast Coloring Oracle for Average Case Hypergraphs, <i>with Edward Pyne, Ronitt Rubinfeld, Asaf Shapira, and Shlomo Tauber</i>, RANDOM (2025), Selected contributed presentation – Highlights of Algorithms 2026, https://arxiv.org/pdf/2507.10691. Characterizing the Distinguishability of Product Distributions through Multicalibration, <i>with Aaron (Louie) Putterman and Salil Vadhan</i>, CCC (2025), https://arxiv.org/pdf/2412.03562. Errors are Robustly Tamed in Cumulative Knowledge Processes, <i>with Anna Brandenberger, Elchanan Mossel, and Madhu Sudan</i>, COLT (2024), Proceedings of the National Academy of Sciences (2025), https://arxiv.org/pdf/2309.05638.pdf.

Uniformity Testing over Hypergrids with Subcube Conditioning, *with Xi Chen*, [SODA \(2024\)](#), <https://arxiv.org/pdf/2302.09013.pdf>.

Uniform scrambles on graphs, *with Lisa Cenek, Lizzie Ferguson, Eyobel Gebre, Jason Meintjes, Ralph Morrison, Liz Ostermeyer, and Shefali Ramakrishna*, [Australasian Journal of Combinatorics \(2023\)](#), <https://arxiv.org/pdf/2108.09821.pdf>.

Preprints

Graph Spectral Sparsification is in Catalytic Logspace, *with Edward Pyne and Ronitt Rubinfeld*, (Submitted) 2026, <https://arxiv.org/pdf/2608.21594>

Quality Control Algorithms for Pattern Counting, *with Madhu Sudan and Ronitt Rubinfeld*, (Submitted) 2026, <https://arxiv.org/pdf/2608.03439>

Recovering Assignments with One-Sided Noise, *with Elchanan Mossel and Colin Sandon*, (Submitted) 2026, <https://arxiv.org/pdf/2607.24073>

The gonality of circulant graphs, *with Lisa Cenek, Lizzie Ferguson, Eyobel Gebre, Jason Meintjes, Ralph Morrison, Liz Ostermeyer, and Shefali Ramakrishna*, (Submitted) 2025, <https://arxiv.org/abs/2508.05761>

Beyond Worst Case Local Computation Algorithms, *with Amartya Shankha Biswas, Ruidi Cao, Edward Pyne, Ronitt Rubinfeld, Asaf Shapira, and Shlomo Tauber*, (Submitted) 2025, <https://arxiv.org/pdf/2403.00129>

Bounds on higher graph gonality, *with Lisa Cenek, Lizzie Ferguson, Eyobel Gebre, Jason Meintjes, Ralph Morrison, Liz Ostermeyer, and Shefali Ramakrishna*, 2022, <https://arxiv.org/pdf/2206.06907.pdf>.

Technical Reports **Data Preparation of the nuMoM2b Dataset ***, *with A. Goretsky, A. Dmitrienko, I. Tang, N. Lari, O. Kunhardt, R. Rashid Khan, A. Catto, D. Mallia, A. Leshchenko, A. Lin, A. Raja, A. Salieb-Aouissi, Itsik Pe'er, R. Wapner, C. Gyamfi-Bannerman*, Technical report, 2021, on medRxiv.

Honors and Awards

- Awarded the **U.S. DoD NDSEG Fellowship** (2024)
- Awarded the **NSF Graduate Research Fellowship**; declined in favor of NDSEG (2024)
- **Magna Cum Laude**, Columbia College (2022)
- **Dean's List**, Columbia College (all applicable semesters)
- National Center for Women & Information Technology: Collegiate Award – **National Finalist** (2021)
- **AWM Student Chapter Award for Scientific Excellence** (2020)
- **Columbia Oxbridge Scholar** – Selected to study Math and CS at St. Anne's College, Oxford. Canceled due to COVID-19. (2019)
- **John Jay Scholar** – Columbia University Scholars Program (2018 - 2022)
- **Verizon App Challenge: Best in Nation**, awarded \$20,000 (2017)

Conference and Invited Talks	· Workshop on Local Algorithms – Poster (2026) · STOC Workshop on Testing in the Modern World (2026) · Tufts Theory Seminar (2026) · Harvard Theory of Computation Seminar – Lightning Talk (2026) · Chicago Junior Theorists Workshop (2025) · MIT Algorithms & Combinatorics Seminar (2025) · MIT Sublinear Reading Group (2025) · Workshop on Local Algorithms (2025) · CCC (2025) · COLT (2024) · SODA (2024) · Young Mathematicians Conference (2021)	
Teaching Experience	· TF for Essential Coding Theory (Harvard CS 229r) Fall 2023 · TA for the PCMI Undergraduate Summer School on Quantum Computation Summer 2023 · TA for Advanced Algorithms (Columbia COMS 4232) Spring 2022 · TA for Introduction to Cryptography (Columbia COMS 4261) Spring 2022 · TA for Analysis of Algorithms I (Columbia CSOR 4231) Summer & Fall 2021 · TA for Artificial Intelligence (Columbia COMS 4701) Spring 2021 · TA for Discrete Mathematics (Columbia COMS 3203) Fall 2020 · Lecturer and TA for Introduction to Proofs Fall 2020, Fall 2021 4-week workshop for undergraduate students that teaches how to write and construct proofs, run by the Columbia Undergraduate Math Society	
Leadership and Seminars	Organizer of TGINF September 2022 - May 2026 TGINF is a seminar for CS theory graduate students and postdocs at Harvard. Founder and Leader of the Columbia Undergraduate Learning Seminar in Theoretical Computer Science March 2021 - May 2022 · Developed ten reading groups, running three groups myself and delegating the organization of the other groups to pairs of graduate and undergraduate students. Co-President of the Columbia-Barnard Association for Women in Mathematics May 2021 - May 2022	
Work Experience	· Microsoft Research – Real-World Reinforcement Learning: Independent Developer in Open Source Fest (Summer 2020) · Google: STEP Software Engineering Intern (Summer 2020) · Amazon: AFE Software Engineering Intern (Summer 2019)	
Reviewing	STOC (2024, 2025), RANDOM (2025, 2026), Applied Probability journals (2025), ITCS (2026), ICALP (2026), FOCS (2026)	